

Chandrayaan-2 Orbiter High Resolution Camera (OHRC) PDS4 Data Products and Archive Software Interface Specification



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DOCUMENT CONTROL AND DATA SHEET

1. Report No. & Date	Chandrayaan-2/DP/SAC/SIPG/HRDPD/TR-05/July 2020
2. Title & subtitle	Chandrayaan-2 – Orbiter High Resolution Camera (OHRC) PDS4 Data Products and Archive Software Interface Specifications
3. Number of Pages	59
4. Number of Reference	11
5. Type of Report	Technical
6. Authors	Data Processing, Payload, Application Team
7. Originating Unit	High Resolution Data Processing Division, Signal and Image Processing Group, Space Applications Centre
8. Abstract	This document provides the Data Products and Archive Software Interface Specifications of OHRC instrument of Chandrayaan-2
9. Key Words	Archive, PDS4, ISDA
10. Security Classification	Open to user community
11. Distribution	To the identified individuals in ISRO/DOS and end users on internet

Chandrayaan-2

Orbiter High Resolution Camera (OHRC)

PDS4 Data Archive

Software Interface Specifications

Document Change History

*A: Addition; M: Modification; D: Deletion

Version No.	Date	Section No.	A/M/D *	Description of change
V 1.0	July 20, 2020	All	A	Initial draft Version
V 2.0	November 03, 2020	6.5	M	Updated the file naming conventions table
V 3.0	November 03, 2020	Appendix-G	A	Added the local data dictionary attributes
V 4.0	July 14, 2021	5.3	M	Updated Data Release and Schedule Plan table by adding the information about the Second Version Release
V 5.0	21 January 2022	5.3	M	Updated Data Release and Schedule Plan table by adding the information about the Third Version Release
V 6.0	19 April 2022	5.3	M	Updated Data Release and Schedule Plan table by adding the information about the Fourth Version Release
V 7.0	28 September 2022	5.3	M	Updated Data Release and Schedule Plan table by adding the information about the Fifth Version Release
V 8.0	October 2023	5.3	M	Updated Data Release and Schedule Plan table
		Appendix-G	M	Added attributes and its value description
		Appendix-H	A	Added the Data Disclaimer
V 9.0	April, 2024	Appendix-H	M	Updated the Data Disclaimer
V 10.0	December 2024	Appendix-G	M	Updated the attributes and its value description
V 11.0	March 27, 2025	5.3	M	Updated Data Release and Schedule Plan table
V 12.0	June 24, 2025	5.3	M	Updated Data Release and Schedule Plan table
V 13.0	23 December 2025	5.3	M	Updated Data Release and Schedule Plan table

Acknowledgements

The authors would like to acknowledge the support and encouragement from Shri DK Das Director SAC for development of Chandrayaan-2 Orbiter High Resolution Camera (OHRC) Data Products Software Interface Specifications & Archive.

Special thanks are due to Shri. Debajyoti Dhar, GD, SIPG, for his interest, constant guidance and useful discussions.

Authors acknowledge to HRDPD members for their support.

Last but not the least we would like to extend our thanks to PDS4 working group members and All the PI's of Chandrayaan-2 for their support.

Technical Content Approvals

		Name & Function
	Reviewed & Approved by	Amitabh DPD (Optical P/L DP) Chandrayaan-2 Shri Aditya Dagar PI, OHRC, Chandrayaan-2

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1. Introduction

1.1 Purpose

The Data Products and Archive software interface specifications describes the data products, format and content of the Orbiter High Resolution Camera (OHRC) Planetary Data System (PDS) data archive bundles in which data products with associated meta data, documentation, and supporting materials are stored. This document is intended to provide enough information to enable users to read and understand the OHRC data products stored in PDS.

1.2 Scope

The specifications in this document apply to all OHRC data products submitted for archival to the ISRO Science Data Archive (ISDA), for all the mission phases of the Chandrayaan-2.

1.3 Data Products and Archive SIS Content Overview

Section 2 of this document describes the Mission overview while Section 3 provides the characteristics of the Orbital High Resolution Camera (OHRC). Section 4 gives an overview of data organization and data flow while Section 5 describes data archive generation, delivery, and validation. Section 6 describes the archive structure and archive production responsibilities. Section 7 describes the file formats used in the archive, including the data product record structures. Individuals involved with generating the archive volumes are listed in Appendix-A While Appendix-B contains sample bundle label, Appendix-C, contains sample calibration collection label Appendix-D contains sample data product label, and Appendix-E contain visualization of PDS4 data products.

1.4 Audience

This document serves both as a SIS and Interface Control Document (ICD). It describes both the archiving procedure and responsibilities, and data archive conventions and format. It is designed to be used both by the instrument data processing team in generating the archive, and by those wishing to understand the format and content of the OHRC PDS data product archive collection. Typically, these individuals would include scientists, data analysts, and software engineers.

1.5 Applicable Documents

- [1] Chandrayaan-2 Sequence of Events (ISRO-URSC-CHANDRAYAAN-2-PR-3171)
- [2] Chandrayaan-2 Configuration Summary Document (ISRO-URSC-CHANDRAYAAN-2-PR-3172)
- [3] Chandrayaan-2 Preliminary Design Review Data Products, Processing and Dissemination Plan - Volume 3A (Optical Payloads) (CHANDRAYAAN-2/DP/SAC/SIPG/HRDPD/TR-04/MAY 2017)
- [4] Preliminary Design Review Document on Orbiter High Resolution Camera (OHRC), (SAC-SEDA-CH2-05-JUL14)
- [5] Chandrayaan-2 Science data management and Archive plan (Chandryaan-2/DP/SAC/SIPG/HRDPD/TR-06/July 2018)
- [6] Chandrayaan-2 SAC DP – ISSDC Interface for Archival (Chandrayaan-2/DP/SAC/SIPG/HRDPD/TR-13/Dec. 2018)
- [7] PDS4 Concepts Data Design Working Group October 1, 2018 version 1.11.0
- [8] Planetary Data System Standards Reference Version 1.11.0 October 1, 2018, Jet Propulsion Laboratory
- [9] The PDS4 Data Provider's Handbook Guide to Archiving Planetary Data Using the PDS4 Standard Version 1.11.0 October 1, 2018
- [10] PDS4 Data Dictionary Abridged – Version 1.11.0.0
- [11] List of Supplemental Formats (Approved by PDS Management Council: 2017-08-30)

1.6 Abbreviations Acronyms, and Description

Following **Table 1** brings out the list of definitions, acronyms & abbreviations used in this document.

Table 1 Acronym, Abbreviations and Description

Acronym	Abbreviations/Description
PDS	Planetary Data System
ISRO	Indian Space Research Organization
ISSDC	Indian Space Data Centre
ISDA	ISRO Science Data archive
SAC	Space Applications Centre
DP	Data Processing
OHRC	Orbiter High Resolution Camera
SPICE	Spacecraft Planet Instrument Camera Event
ICD	Interface Control Document
Kernel	Primary SPICE data sets
DPD	Deputy Project Director
Archive Team	Collection of teams from science (PI), Payload, Data Processing and ISSDC Operations

1.7 Glossary

Table 2 shows glossary of terms used in PDS4 domain.

Table 2 PDS4 Glossary

Terms	Definition
Archive	A place in which public records or historical documents are preserved; also the material preserved – often used in plural. The term may be capitalized when referring to all of PDS holdings – the PDS Archive.
Basic Product	The simplest product in PDS4; one or more data objects (and their description objects), which constitute (typically) a single observation, document, etc. The only PDS4 products that are not basic products are collection and bundle products
Bundle Product	A list of related collections. For example, a bundle could list a collection of raw data obtained by an instrument during its mission lifetime, a collection of the calibration products associated with the instrument, and a collection of all documentation relevant to the first two collections.
Class	The set of attributes (including a name and identifier) which

	describes an item defined in the PDS Information Model. A class is generic – a template from which individual items may be constructed.
Collection Product	A list of closely related basic products of a single type (e.g. observational data, browse, documents, etc.). A collection is itself a product (because it is simply a list, with its label), but it is not a basic product.
Data Object	A generic term for an object that is described by a description object. Data objects include both digital and non-digital objects.
Description Object	An object that describes another object. As appropriate, it will have structural and descriptive components. In PDS4 a 'description object' is a digital object – a string of bits with a predefined structure.
Digital Object	An object which consists of real electronically stored (digital) data.
Identifier	A unique character string by which a product, object, or other entity may be identified and located. Identifiers can be global, in which case they are unique across all of PDS (and its federation partners). A local identifier must be unique within a label.
Label	The aggregation of one or more description objects such that the aggregation describes a single PDS product. In the PDS4 implementation, labels are constructed using XML.<say what it contains>
Logical Identifier (LID)	An identifier which identifies the set of all versions of a product.
Versioned Logical Identifier (LIDVID)	The concatenation of a logical identifier with a version identifier, providing a unique identifier for each version of product.
Manifest	A list of contents.
Metadata	Data about data – for example, a 'description object' contains information (metadata) about an 'object.'
Non-Digital Object	An object which does not consist of digital data. Non-digital objects include both physical objects like instruments, spacecraft, and planets, and non-physical objects like missions, and institutions. Non-digital objects are labeled in PDS in order to define a unique identifier (LID) by which they may be referenced across the system.
Object	A single instance of a class defined in the PDS Information Model.
PDS Information Model	The set of rules governing the structure and content of PDS metadata. While the Information Model (IM) has been implemented in XML for PDS4, the model itself is implementation independent.
Product	One or more tagged objects (digital, non-digital, or both) grouped together and having a single PDS-unique identifier. In the PDS4

	implementation, the descriptions are combined into a single XML label. Although it may be possible to locate individual objects within PDS (and to find specific bit strings within digital objects), PDS4 defines 'products' to be the smallest granular unit of addressable data within its complete holdings.
Tagged Object	An entity categorized by the PDS Information Model, and described by a PDS label.
Registry	A data base that provides services for sharing content and metadata.
Repository	A place, room, or container where something is deposited or stored (often for safety).
XML	eXtensible Markup Language.
XML schema	The definition of an XML document, specifying required and optional XML elements, their order, and parent-child relationships.

2. Mission Overview

Chandrayaan-2, the second Indian mission to the moon was launched on 22nd July 2019 with Orbiter, Lander and Rover configuration. The mission is a highly complex mission, which represents a significant technological leap compared to the previous missions of ISRO. It comprised an Orbiter, Lander and Rover to explore the unexplored South Pole of the Moon. The mission is designed to expand the lunar scientific knowledge through detailed study of topography, seismography, mineral identification and distribution, surface chemical composition, thermo-physical characteristics of top soil and composition of the tenuous lunar atmosphere, leading to a new understanding of the origin and evolution of the Moon. After the injection of Chandrayaan-2, a series of maneuvers were carried out to raise its orbit and on August 14, 2019, following Trans Lunar Insertion (TLI) maneuver, the spacecraft escaped from orbiting the earth and followed a path that took it to the vicinity of the Moon. On August 20, 2019, Chandrayaan-2 was successfully inserted into lunar orbit. While orbiting the moon in a 100 km lunar polar orbit, on September 02, 2019, Vikram Lander was separated from the Orbiter in preparation for landing. Subsequently, two de-orbit maneuvers were performed on Vikram Lander so as to change its orbit and begin circling the moon in a 100 km x 35 km orbit. Vikram Lander descent was as planned and normal performance was observed up to an altitude of 2.1 km. Subsequently communication from lander to the ground stations was lost. The Orbiter placed in its intended orbit around the Moon will enrich our understanding of the moon's evolution and mapping of the minerals and water molecules in Polar regions, using its eight state-of-the-art scientific instruments. OHRC is the highest resolution camera (0.25 m) in any lunar mission so far and will provide high resolution images which will be immensely useful to the global scientific community. The precise launch and mission management has ensured a long life of almost seven years instead of the planned one year. There are broadly two type of objectives of OHRC namely 'mission objectives' and 'science objectives'. Mission objective is prime for the safe landing of the Chandrayaan-2 lander-craft by making use of the OHRC images and derived datasets like crater catalogue and hazard map (for boulders, craters, slopes and shadows).

2.1 Mission Objective

OHRC's mission objective is to capture images of the Moon surface with high spatial resolution to characterize the landing site (prior to landing) with a GSD better than 0.32 m during two oblique views (0.25 m at nadir) from 100 km altitude. Nominal OHRC imaging region of landing site region is about 3 km × 12 km area in two orbits with different view angles (with spacecraft maneuvering) to generate digital elevation model (DEM) information. Major hazards for lander-craft are slopes, boulders, craters and shadows for which OHRC data is important. Slope from DEM and shadows from brightness images are required to generate the hazard map and safe-grid location. To expand the lunar scientific knowledge through detailed study of topography,

mineralogy, surface chemical composition, thermo-physical characteristics and tenuous lunar atmosphere leading to a better understanding of the origin and evolution of the Moon.

2.2 Payload Objective

OHRC's science objectives are supported by its very high spatial resolution imaging capability. Also low illumination imaging capability adds values to the observations in the near polar regions. It can image small-scale features in fine detail. The first objective of OHRC, after mission objective, i.e. the landing-site imaging, is to image the anthropogenic sites on lunar surface. The other scientific objectives are: (i) to study the recent volcanism and other small-scale volcanic sites; (ii) to study the central peaks of young impact craters, and (iii) to understand tectonic (seismic) processes (through boulder population determination, boulder rolling trails, seismicity related features, etc.); impact cratering (small craters, ejecta material, melt pools, etc.) and mass wasting (rock falls, debris flow, etc.).

3. Instrument Overview

OHRC used to image the moon surface with high spatial resolution. OHRC images used to characterize the landing site (as its primary goal) with a ground sampling of 0.25-0.32 m. This data is used to identify and characterize the landing area before the landing of Lander. OHRC images used to identify the major hazards of importance like slopes, boulders, craters and shadows. Based on these observations, the landing path (trajectory) is planned. Currently OHRC is used for imaging scientific regions of interest (as its secondary goal) and identification and characterization of future landing sites.

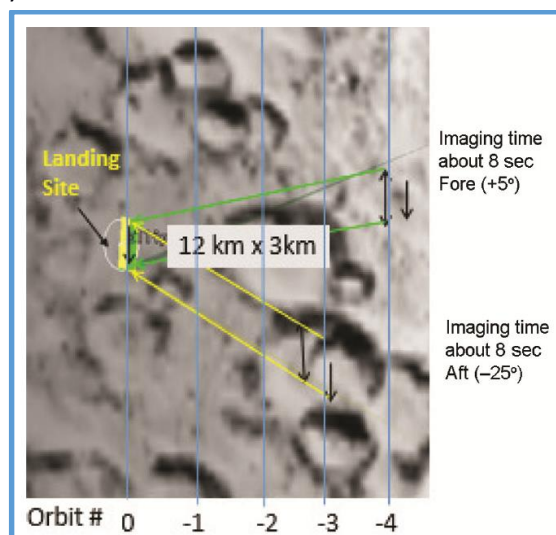


Figure 1 OHRC principle of operation

This camera is based on reflective system with large aperture of about 300 mm and 12K TDI detector. The camera generates monochrome images with ground sampling of 0.25 m from 100 km altitude and swath of about 3 km at nadir. **Table 3** shows the OHRC Payload features.

Table 3 OHRC Payload Features

Parameter	Values
Orbit altitude (km)	100
GSD (m) at nadir	0.25
Swath (km) at nadir	3
Spectral range (nm)	450–800
Telescope diameter (mm)	300
Detector	12 K by 256 TDI
Quantization (bits)	10 (electronics) 8 (transmission)
Reference illumination condition	8% Albedo, 5–6° Sun elevation
Reference radiance (mW/cm ² /sr/μm)	0.5
Saturation radiance (mW/cm ² /sr/μm)	0.8 with 256 TDI
Signal-to-noise ratio (SNR) @ reference radiance	100 with 256 TDI (140 at saturation)
Stereo views	Fore and Aft in two consecutive orbits by spacecraft and maneuvering

With 0.25m GSD from 100km altitude, dwell time is around 160μsec and very low photon signal has to be detected by OHRC with SNR requirement of greater than 50.

Time Delay Integration (TDI) based imaging sensor has an advantage over external frame addition technique in terms of read noise components

3.1 Camera Configuration

Orbiter High Resolution Camera (OHRC) is mounted on the negative pitch panel of the Orbiter spacecraft. The OHRC data will be transmitted to ground for characterizing the landing site and generating DEM.

3.2 Optical Configuration

Optical layout is worked out to meet the system requirements. The basic configuration of OHRC telescope is based on the modified RC design. The primary and secondary mirrors of the telescope are hyperbolic in surface profile followed by an FCO (field correcting optics). The optical schematic of OHRC is shown in figure 2.

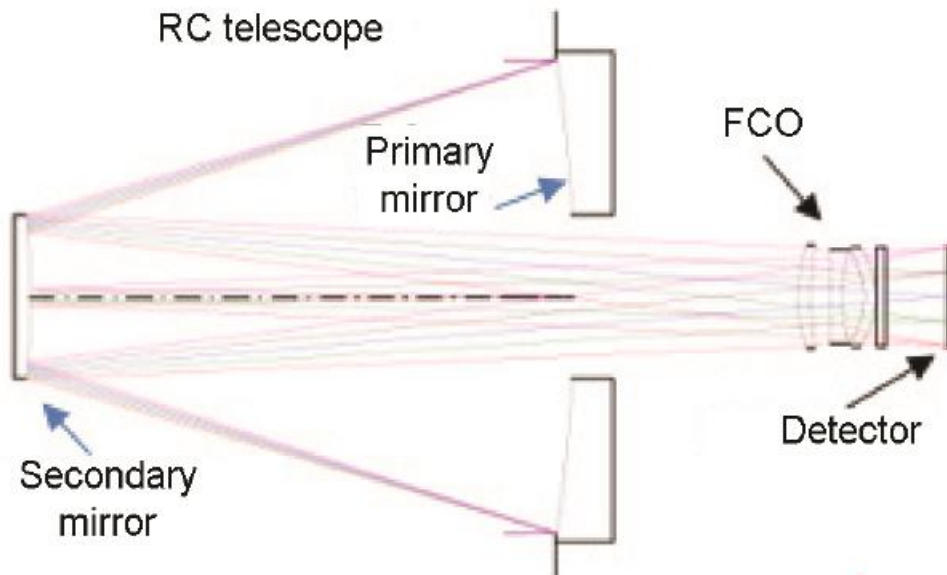


Figure 2 OHRC optical layout

Parameter	Value
Optical configuration	Ritchey–Chretien telescope with field correcting optics (RC + FCO)
Spectral range (nm)	450–800
Primary mirror diameter, D (mm)	300
Effective focal length (EFL; mm)	2046
Field-of-view (FOV, deg)	± 0.86
MTF (%) @ 96 lp/mm (Nyquist)	24

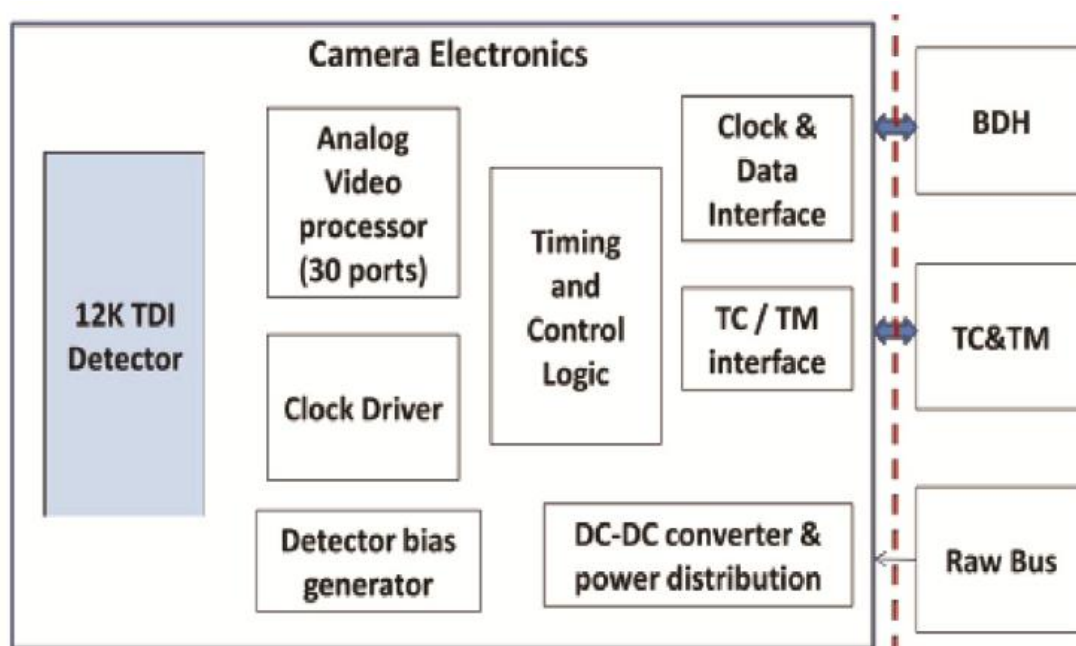
3.3 Detector Configuration

Detector parameters are tabulated below.

Parameter	Value
Detector type	TDI-CCD
Pixel size (μm)	5.2×5.2
Spectral range (nm)	450–800
TDI stages	256, 192, 128, 64 (selectable)
Full well capacity (ke^-)	26.6
Read noise (e^-)	40

3.4 Electronics Configuration

Electronics block diagram is shown below.



Electronics parameters are tabulated below.

Parameter	Value
Quantization (bit)	10 (electronics) 8 (transmission) by bit selection
Integration times (μs)	7 values (selectable by telecommand (TC)) 213.84, 209.58, 205.32, 181.74, 178.14, 174.87 (for landing site imaging catering to ± 2 km altitude variation, fore and aft view) and 162.10 μs (for Nadir view imaging)
TDI stages	256, 192, 128, 64 (selectable by TC) 256 (for landing site) 64 (post-landing)
Data volume (per view)	4.6 Gb for one $3 \text{ km} \times 12 \text{ km}$ strip (for 8 bits)
Raw power (W)	~ 36

4. Data Overview

This section provides a high level description of archive organization under the PDS4 Information Model (IM) as well as the flow of the data from the spacecraft through delivery to PDS. Unless specified elsewhere in this document, the Chandrayaan-2 OHRC archive confirms with version 1.11.0.0 of the PDS4 IM [4] and version 1.11 of the Chandrayaan-2 mission schema. A list of the XML Schema and Schematron documents associated with this archive are provided in **Table 4** below.

Table 4 Chandrayaan-2 OHRC Archive Schema and Schematron

XML Document	Steward	Product LID
PDS Master Schema, v.1.0.0.0	ISDA	urn:isro:isda:system_bundle:xml_schema:pds-xml_schema

4.1 Data Processing Levels

A number of different systems may be used to describe data processing level. This document refers to data by their PDS4 reduction level. Three different types of data products are envisaged for OHRC payload of Chandrayaan-2. **Table 5** provides a description of these levels along with the equivalent designations used in other systems.

Table 5 Chandrayaan-2 OHRC Data Processing Levels

PDS4 Level	PDS4 Level Description	ISRO Level	CODMAC Level
Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format.	0	2
Calibrated	Data converted to physical units, which makes values independent of the instrument.	1	3

4.2 Data Products

A PDS product consists of one or more digital and/or non-digital objects, and an accompanying PDS label file. Labeled digital objects are data products (i.e.

electronically stored files). Labeled non-digital objects are physical and conceptual entities which have been described by a PDS label. PDS labels provide identification and description information for labeled objects. The PDS label defines a Logical Identifier (LID) by which any PDS labeled product is referenced throughout the system. In PDS4 labels are XML formatted ASCII files. More information on the formatting of PDS labels is provided in Section 6.3. More information on the usage of LIDs and the formation of Chandrayaan-2 LIDs is provided in Section 6.1. The PDS4 Data Products for OHRC defined based on the defined data processing levels (L0 and L1) as Raw and Calibrated data products. Table 6 shows the data products definitions for each levels of data processing. To visualize the data products please refer Annexure E. more details shown in next sub sections.

Table 6 Chandrayaan-2 OHRC Data Products Definitions

Data Products Levels	Level-0	Level-1
Data Products Definition	Raw data without any data processing correction along with system level corner coordinates	Radiometrically corrected, refined corner coordinates tagged and grid data
PDS4 Level	Raw	Calibrated

4.2.1 Raw Data Products

For each payload observation, PDS4 data products are generated (based on product id & sensor id derived from file name (section 6.5)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. For example, user has downloaded the zip product as

- ch2_ohr_nrp_20200229T0938004033_d_img_d32.zip

Figure 3 shows unzipped structure of OHRC Raw Data Product zip file at user end.

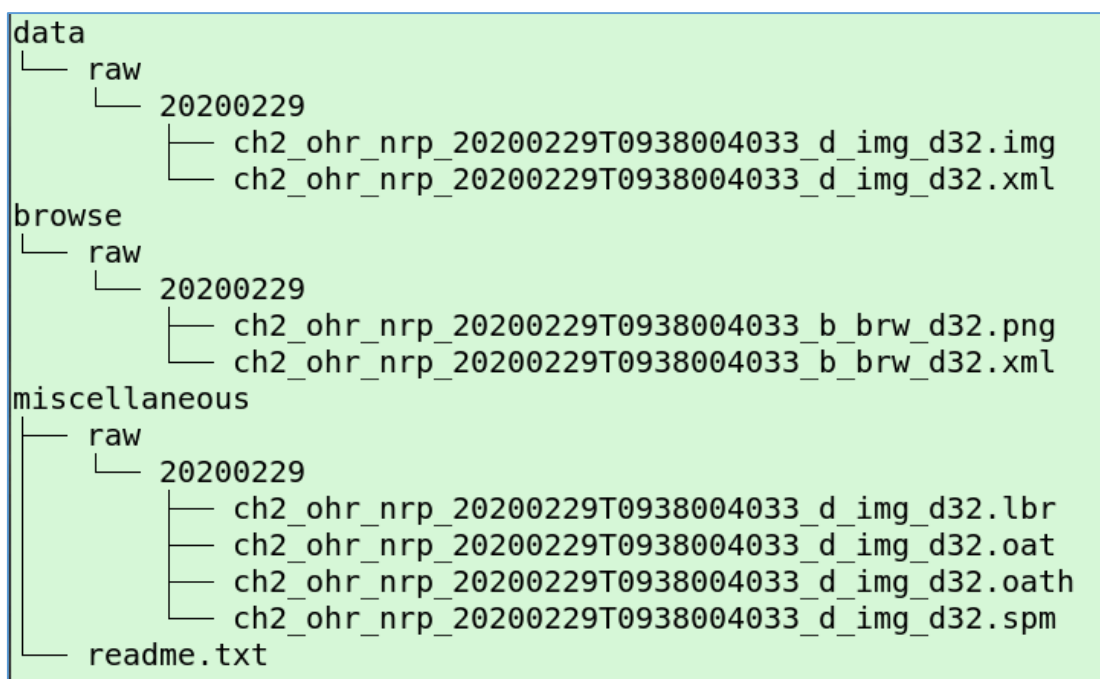


Figure 3 OHRC Raw Data Archive

Raw Data Products contains raw data without any correction along with the system level corner coordinates. **Table 7** shows the details of individual data products (with the data formats) present inside the zip product.

Table 7 OHRC Raw Data Products

PDS4 Data Product Levels	PDS4 Data Products	Data Formats	Description	Visualization tools available
Raw	Data Raw Image	Binary	The PDS Raw data in generic binary	Any binary image viewer – imageJ, envi, erdas by providing the details of image width, image height and data type from label file
	Data Raw Label	XML	The PDS Label Product contains description about the Raw Product.	PDS4 Viewer
	Browse Raw Image	PNG	Sub sampled browse raw image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Raw Label	XML	The PDS Label Product contains description about the Browse Raw Product.	

	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data
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4.2.2 Calibrated Data Products

For each payload observation, PDS4 data products are generated (based on product id derived from file name (section 6.5)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. For example, user has downloaded the zip product as

- ch2_ohr_ncp_20200229T0938004033_d_img_d32.zip

Figure 4 shows unzipped structure of OHRC Calibrated Data Product zip file at user end.

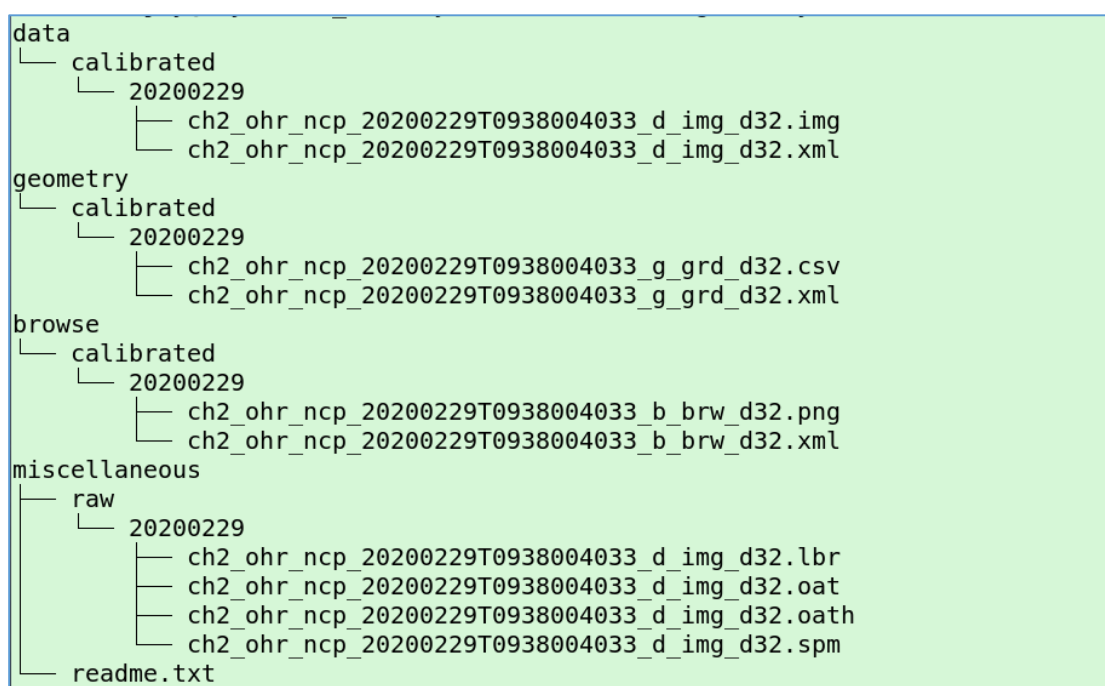


Figure 4 OHRC Calibrated Data Archive

Calibrated Data Products contains radiometrically corrected data along with the refined corner coordinates.

Table 8 shows the details of individual data products (with the data formats) present inside the zip product.

Table 8 OHRC Calibrated Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
Calibrated	Data Calibrated Image	Binary	The PDS Radiometrically corrected data in generic binary format	Any binary image viewer – imageJ, envi, erdas by providing the details of image width, image height and data type from label file
	Data Calibrated Label	XML	The PDS Label Product contains description about the Calibrated Product.	PDS4 Viewer
	Browse Calibrated Image	PNG	Sub sampled browse calibrated image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Calibrated Label	XML	The PDS Label Product contains description about the Browse Calibrated Product.	
	Geometric Calibrated Grid	CSV	A grid file containing lon, lat, scan and pix as four columns along with the data	Any text editor, ms excel can be used to load the csv file.
	Geometric Calibrated Grid Label	XML	The PDS Label Product contains description about the Geometry Calibrated Product.	PDS4 Viewer
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

4.3 Data Products Bundle

The OHRC data archive is organized into 2 bundles at data level. A description of each bundle is provided in **Table 9**. A more detailed description of the contents and format

of each bundle is provided in Section 6.2.

Table 9 OHRC Data Products Bundles

Bundle Logical Identifier	PDS4 Reduction Level	Description	Data Provider
urn:isro:isda:ch2.ohr.raw	Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format	ISSDC
urn:isro:isda:ch2.ohr.calibrated	Calibrated	Data converted to physical units, which makes values independent of the instrument.	ISSDC

4.4 Data Flow

This section describes only those portions of the Chandrayaan-2 data flow at ISSDC that are directly connected to archiving. The ISSDC will maintain an active archive of all Chandrayaan-2 science data, and will provide the PI with direct access through the life of the Chandrayaan-2 mission. After the end of the Chandrayaan-2 project, ISDA will be the sole long-term archive for all public Chandrayaan-2 data.

5. Archive Generation

The OHRC archive products are produced by the SAC DP team in cooperation with the ISSDC. The archive volume creation process will be described in this section. Archived data established at ISSDC from the SAC DP software running at ISSDC, are made available to users electronically as soon as it qualifies as a science product but not later than nine months after the delivery and validation of the data.

5.1 Data Processing and Production Software

After receiving Level 0 data from the level-0 software at ISSDC, the ISSDC will process the Level 0 data into science data products using software provided by the SAC DP. The ISSDC will provide the Level 0 data files with all the ancillary data required for science processing to the SAC DP Software. From this data, the SAC DP Software will generate Level 0/1 raw and calibrated science data products. The science data products that the SAC DP Software delivers to the ISSDC will be stored by the ISSDC for the duration of the project, and will be made available to the PI team or as per the ISRO data distribution policy. The ISSDC will deliver archival-quality science data products to the ISDA for distribution to the public and long-term archiving in accordance with the OHRC Data Products & Archive SIS (this document).

5.2 Data Validation

There are two types of data validation. The first is validation of the science data and the second one is validation of the compliance of the archive with PDS archiving and distribution requirements. The first type of validation is carried out by the PI Team, and the second is overseen by the ISDA, in coordination with the PI Team. The formal validation of data content, adequacy of documentation, and adherence to PDS archiving and distribution requirements is scheduled and coordinated by the ISDA. This review includes peer review from the science community, at the ISDA's discretion. The review process may result in "liens," which are actions that are recommended by the reviewers or by PDS personnel to correct the archive. All liens must be resolved by the data-set provider. Once the liens are cleared, the ISDA completes a final validation before release.

5.3 Data Transfer Methods and Delivery Schedule

The Data Processing & Archival Software is running at ISSDC during each mission phases of Chandrayaan-2. The products are generated at ISSDC. Initially active archive is established data will be collected as per mission operation plan. The products are available for the PI as initial validation of data. Table 10 shows OHRC data release schedule plan for Chandrayaan-2 mission.

Table 10 Data Release Schedule Plan

Payload	Release*	Data Collection Period (from date to date)	Lock-in-period (as per SSPO ISRO HQ)	Date Released to Public**	Products
OHRC	First Version	February (28 and 29) & August (24)	9 month	December 2020	Raw and Calibrated
	Second Version	May 2020 to September 2020	9 month	July 2021	Raw and Calibrated
	Third Version	March 2021 to April 2021	9 month	Feb. 2022	Raw and Calibrated
	Fourth Version	Oct-Dec 2021 Jan-Mar2022	6 month	April 2022	Raw and Calibrated
	Fifth Version	06 Sept 2019 28-29Jan 2022	6 month	Sept. 2022	Raw and Calibrated
	Sixth Version	Dec. 2021 & (Feb-Mar-May-June-Aug.) 2023	-	October 2023	Raw and Calibrated
	Seventh Version	Sept. and Oct. 2023 & Feb, Mar, Apr 2024	-	April 2024	Raw and Calibrated
	Eighth Version	Apr 2024	-	May 2024	Raw and Calibrated
	Ninth Version	Aug. 2024 to Mar. 2025	-	April 2025	Raw and Calibrated
	Tenth Version	Sept. 2024 to Mar. 2025	-	June 2025	Raw and Calibrated
	Eleventh Version	Sept. 2025 to Nov. 2025		Dec 2025	Raw and Calibrated

* Archive for the duration of Sept. 2025 to Nov 2025 will be released to the user. *** Release dates are dates that the ISDA releases Chandrayaan-2 archives to the public. Chandrayaan-2 delivers archives to the ISDA at least two weeks before the date of release to the public.

5.4 Data Product and Archive Volume Size Estimates

The total Archive Volume Size is worked out based on the data size, no. of orbits per day, mission life.

5.5 Backups and duplicates

The ISDA is responsible for maintaining at least three copies of its science archives and for delivering one copy of the data to the ISSDC for archive. As archives are released, ISSDC generate at least three copies on appropriate physical media for long-term storage.

6. Archive Organization and File Naming Conventions

6.1 Namespace Registration

PDS4 archive standard is adopted for Chandrayaan-2 mission. PDS4 is international archive standard being adopted by various space agencies across globe. There is International Planetary Data Alliance (IPDA), where all space agencies are members, ISRO is also one of them. IPDA has adopted PDS4. First step towards starting archive activities for any space agency is to register namespace via IPDA. ISRO has submitted namespace to IPDA. The namespace will be used as root to the archive of ISRO's upcoming and future planetary missions. For Chandrayaan-2 mission following name space will be used which is shown in the **Table 11**.

Table 11 ISDA Namespace for PDS4

Namespace ID	Logical Identifier prefix	Authority	Steward	Steward_ID	Provider/contact Point
isda	urn:isro:isda	ISRO Science Data Archive	ISRO	isda	Ajay Kumar Prashar T P Srinivasan B.N. Ramakrishna

6.2 Logical Identifiers

Logical Identifier(LID) is unique to the product. Versioned Identifier (VID) indicates the versions improvements with respect to the products. All LIDs consist of a series of colon-separated segments.

Namespace:{BundleID}_{SubBundleID}:{CollectionID}_PDS4DataProductLevel:[{ProductID}]]. **Table 12** shows values for LID construction.

Table 12 LID for Chandrayaan-2 mission

Namespace	Bundle Id	Sub Bundle Id	Collection Id	PDS4 Data Product Level	Product Id
urn:isro:isda	ch2	cho	ohr	raw calibrated	Address Data Products SIS document or EAICD

The following **table 13** shows the construction of LID to be used for all the scientific payloads of Chandrayaan-2 mission.

Table 13 LID for Chandrayaan-2 mission instruments

Instrument id	Logical Identifier
ohr	urn:isro:isda:ch2_cho.ohr:data_raw:productId urn:isro:isda:ch2_cho.ohr:data_calibrated:productId

Products, collections, and bundles may evolve over time as calibration or other processing changes lead to improved versions. PDS appends a VID (version Identifiers) of the form M.n to LID to indicate different versions. The suffix is separated from LID by double colon. Archive structure is designed both at mission phase level and instrument level.

6.3 Mission Archive Structure at Bundle Level

At mission level, data is organized as mission id wise bundle. The mission id wise bundle at next level is organized based on the instrument host id wise bundle. The instrument host id bundle at next level is organized based on the mission phase id. Each mission phase id is having instrument id wise collection. All these mission attributes - mission id, instrument host id, mission phase id and instrument id are defined under archive conventions. For OHRC instrument, **table 14** shows values defined for mission attributes in archive conventions.

Table 14 Mission Attributes defined at structure level

Mission Attributes	Values	Defined at structure level
mission id	ch2	ch2_bundle
instrument host id	cho	cho_bundle
mission phase id	nop	nop (normal operation phase)
instrument id	ohr	ohr_collection

All these attributes mentioned in above table are used in designing the archive structure at mission level. The figure 5 shows the mission level data products archive organization.

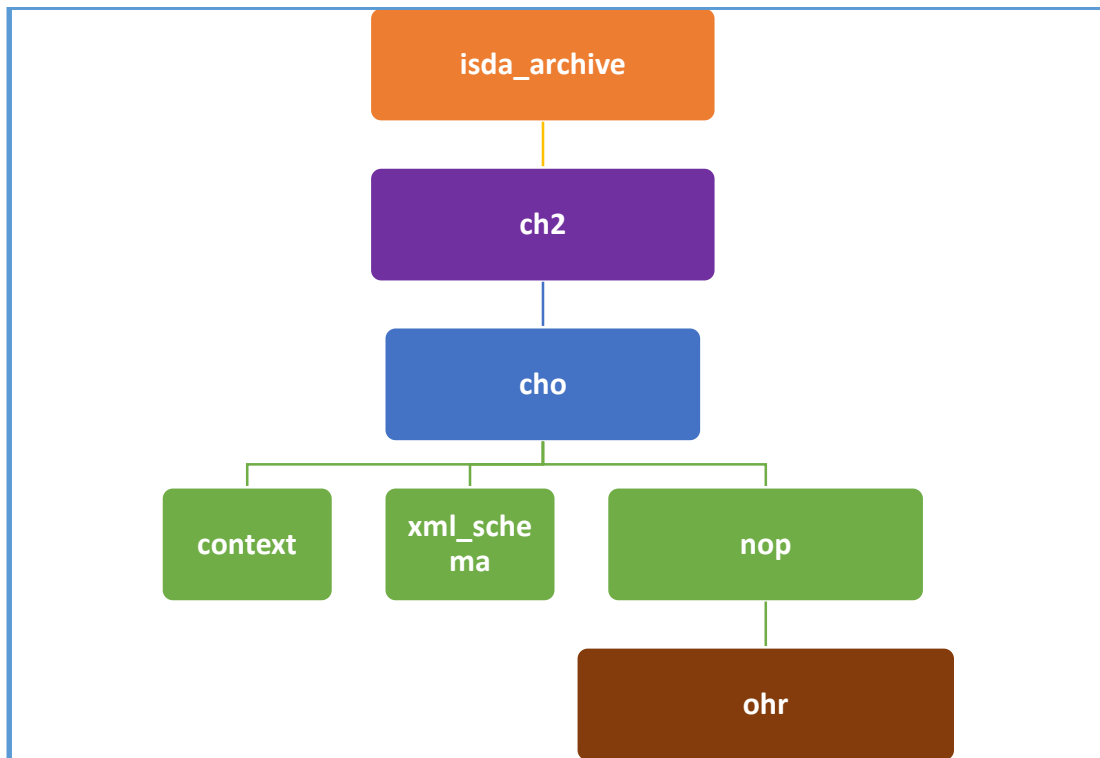


Figure 5 Chandrayaan-2 Mission Level Archive Organization

6.4 Instruments Collection

Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions.

To handle requirement of archiving the data products for 12 orbits in a day, data products is organized into year, month and day wise as `yyyymmdd`.

The year month and day wise directory contains PDS4 data products including PDS4 Label products. Figures 6 shows the detail of OHRC instrument wise collection structure.

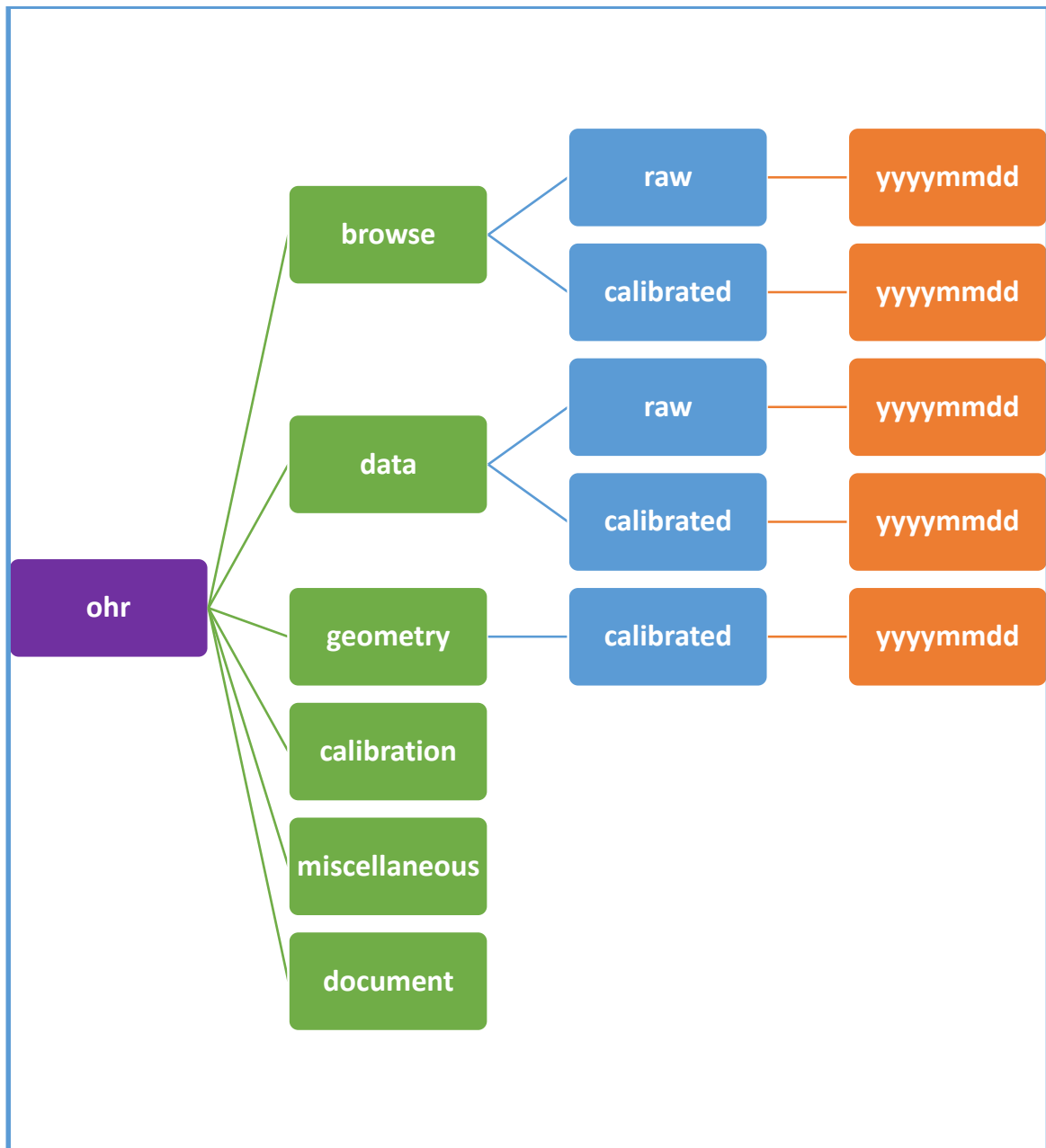


Figure 6 Chandrayaan-2 OHRC Instrument wise collection structure

At instrument level, collection of different product types is defined as per standard PDS4 conventions. Table 15 defined in detail about collection types.

Table 15 Collection product types

File or Directory Name	Description
root	Root directory describing full ohrc collection data sets
browse	Contains 'quick-look' products designed to facilitate use of the archive
calibration	Contains data and files necessary for the calibration of basic

	products.
context	Contains list of products comprising various objects, identified with in the PDS4 registry, that are specific to the science bundle. These include physical objects such as instruments, spacecraft, and planets and conceptual objects such as missions.
document	Contains all components (figure, tables, text, etc.) of one or more related documents. The document include software interface specifications (SISs), calibration reports.
data	Contains observational products, often separated according to processing level, target, instrument mode, etc.
xml_schema	Contains XML schemas and related products which may be used for generating and validating PDS4 labels.
geometry	Contains non-SPICE geometry products
miscellaneous	Contains supplementary information provided by data provider to be useful in the interpretation and use of other collections in the bundle but which does not fit within the scope of the other collections. for example meta data catalogs, database dumps, and records of modification history could be included.

6.4.1 Root directory content

Files in the root directory include an overview of the archive, a description of the collection for the PDS4 data set. The following files are contained in the Root Directory shown in Table 16.

Table 16 Root directory content products

File Name	File Contents	File Provided By
bundle.xml	Collection content and format information	Archive Team
readme.txt	Collection content and format information	Archive Team

6.4.2 Browse directory content

Under the Browse directory there is a separate subdirectory which is year, month and day wise for each type of product. The year, month and day wise subdirectory contains browse image for each image, which is a down sampled version of the full resolution image, with the size of m pixels x n lines and 8 bits per pixel, is defined as an object and contained in the data file. PNG images derived from the browse image are stored

in this directory, as well as the label files pointing to these images. All the browse PNG images are named as per file naming convention.

6.4.3 Calibration directory content

The calibration directory contains calibration files used to process the data products, or calibration data needed to use the data products. This calibration files are the look up table files that got generated during ground calibration exercise. All these lookup table files naming convention is based on the parameters – mission (ch2), instrument_id (ohr) byte selection (lsb/mid/lsb) and TDI stages (TDI64/TDI128/TDI256/TDI192). The following files are contained in the calibration directory shown in **table 17**.

Table 17 Calibration directory content products

File Name	File Contents	File Provided By
collection_calibration_inventory.csv	Contains about the list of calibration data products in the collection inventory	Archive Team
collection_calibration_inventory.xml	XML Label defining the inventory csv file	Archive Team
Calibration files	Look up table files	Archive Team
Calibration xml files	XML label defining the look up table	Archive Team

6.4.4 Context directory content

The context directory contains the list of products comprising various objects, identified with in the PDS4 registry, that are specific to the science bundle. These include physical objects such as instruments, spacecraft, and planets and conceptual objects such as missions. The following files are contained in the context directory shown in **table 18**.

Table 18 Context directory content products

File Name	File Contents	File Provided By
collection_context_inventory.csv	Contains about the list of context data products in the collection inventory	Archive Team
collection_context_inventory.xml	XML Label defining the inventory csv file	Archive Team

6.4.5 Document directory content

The document directory contains documentation to help the user understand and use the archive data. The following files are contained in the document directory shown in the **table 19**.

Table 19 Document directory content products

File Name	File Contents	File Provided By
collection_document_inventory.xml	Label to define the inventory	Archive Team
collection_document_inventory.csv	A description of the contents of this directory	Archive Team
ch2_ohrc_pds_dp_archive_sis.pdf	Description of instrument and PDS data products and Archive Structure	Archive Team
ch2_ohrc_pds_dp_archive_sis.xml	XML label defining ohrc pds sis pdf	Archive Team

6.4.6 Data directory content

Under the data directory there are three separate subdirectory based on the defined PDS4 data products levels – raw, calibrated and derived. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains PDS4 data products mainly image and associated label files in XML format. All the browse binary images are named as per file naming conventions mentioned under section 6.5.

6.4.7 Xml_schema directory content

This directory contains XML schemas and related products which may be used for generating and validating PDS4 labels. The following files are contained in the xml_schema directory shown in **table 20**.

Table 20 Xml_schema directory content products

File Name	File Contents	File Provided By
collection_xml_schema_inventory.csv	Contains about the list of xml schema in the collection inventory	Archive Team
collection_xml_schema_inventory.xml	XML Label defining the xml schema csv file	Archive Team

6.4.8 Geometry directory content

Under the geometry directory there are three separate subdirectory based on the defined PDS4 data products levels – raw, calibrated and derived. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains PDS4 data products mainly a grid file in csv format and associated label files in XML format. All the grid files are named as per file naming conventions in section 6.5.

6.4.9 Miscellaneous directory content

Under the miscellaneous directory there are two separate subdirectory based on the defined PDS4 data products levels – raw and calibrated. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains data products mainly orbit and attitude file (oat), orbit and attitude header file (oath), liberation angle file (lbr) and sun parameter file (spm) in ASCII text format All the files with extension oat, oath, lbr and spm files are named as per file naming conventions in section 6.5. It also contains the supplementary information provided by data provider to be useful in the interpretation. The following files are contained in the Document Directory shown in **table 21**.

Table 21 Miscellaneous directory content

File Name	File Contents	File Provided By
collection_miscellaneous_inventory.csv	Contains about the list of miscellaneous in the collection inventory	Archive Team
collection_miscellaneous_inventory.xml	XML Label defining the xml schema csv file	Archive Team

6.5 File Naming Conventions & Formats

This section describes the naming convention used for science data files for the Chandrayaan-2 mission. **Table 22** shows the detail description about the following file naming conventions designed for the data products.

ch2_<inst>_<mtc>_<YYYYMMDDTHHMMSSsss>_<p>_<prd>_<stn>.fff

Table 22 File Naming Conventions

Code	Description
ch2	It denotes the Mission name whose value is fixed as ch2
inst	It denotes the Instrument id, it can have following value = tmc (refers to the TMC2) = iir (refers to the IIRS) = ohr (refers to the OHRC)
m	It denotes the mission phase name, it can have following value = e (refers to the Earth Bound Phase - ebp) = n (refers to the Normal Operations Phase - nop)
t	It denotes the data type; it can have following values: = r (it denotes raw data) = c (It denotes calibrated data) = d (It denotes derived data)
c	It denotes the imaging mode/camera id/band. It can be one of the following: = p (it denotes panchromatic camera)
YYYYMMDDTHHMMSSsss	It denotes the observation/imaging start time in format [YYYYMMDDTHHMMSSsss] YYYY = stand for year MM = stand for month in MM format DD = stand for day in DD format T = stand for time. HH = stand for hour in HH format MM = stand for minute in MM format SSS = stand for seconds in SS format sss = stand for 1/10 millisecond or 100 microsecond in ssss format
P	It denotes the PDS Data Products categories. It value can be any of the following: = d (It denotes data products under data directory)

	= b (it denotes browse data products under browse directory) = g (it denotes gridded data products under geometry directory)
Prd	It denotes the PDS Data Product Name. It value can be any of the following: = img (It denotes the image data product) = brw (It denotes the browse data product) = grd (It denotes the gridded data product) = dtm (it denotes DTM derived product) = orh (it denotes Ortho derived product)
Stn	It denotes the station id from which data was acquired. It value can be any of the following: = d32 (ISSDC Bangalore) = d18 (ISSDC Bangalore) = gds (Gold Stone USA) = cnb (Canberra, Australia)
fff	It denotes the standard file type extension. It can have any one of the following values: = img (PDS Image Data File) = xml (PDS Detached Label File) = jpg (PDS Browse Data File) = csv (PDS Geometry Data File) = tif (PDS Geo tiff data for derived products)

7. Archive Products formats

Data that comprises the OHRC archives are formatted in accordance with PDS specifications. This section provides details on the formats used for each of the products included in the archive. The detailed file formats for the defined data products based on PDS4 data types have been defined shown in next sub section.

7.1 Data File Formats

OHRC instrument is an imaging payload. The primary data collected from instrument is in image format. The primary data collected from instrument is raw image. It is a panchromatic camera. Both primary as well as secondary data sets are defined that contains data product levels, types and formats shown in the **table 23**.

Table 23 OHRC PDS4 Data Products

Sensors	PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Data type*
Panchromatic	Raw	Image	Binary	UnsignedByte
	Raw	Browse	PNG	UnsignedByte
	Calibrated	Image	Binary	UnsignedByte
	Calibrated	Browse	PNG	UnsignedByte
	Calibrated	Geometry	CSV	ASCII Text

Note*:

UnsignedLSB2 is UInt16 (Unsigned Integer of 16-bit length)

SignedLSB2 is Int16 (Signed Integer of 16-bit length)

Unsigned Byte is 8 bit

7.2 Document Product File Formats

All the files under document collection will be written in text format, csv, png, xml and pdf

7.3 PDS Labels

PDS labels are ASCII text files written, in the eXtensible Markup Language (XML). All product labels are detached from the digital files (if any) containing the data objects they describe (except Product_Bundle). There is one label for every product. Each product, however, may contain one more data objects. The data objects of a given

product may all reside in a single file, or they may be stored in multiple separate files. PDS4 label files must end with the file extension “.xml”.

7.3.1 XML Documents

For the Chandrayaan-2 mission PDS labels will conform to the PDS master based schema based upon the 1.11.0.0 version of the PDS Information Model for structure, and the 1.11.0.0 version of the PDS schematron for content. By use of an XML editor these documents may be used to validate the structure and content of the product labels.

The PDS master schema and schematron documents are produced, managed, and supplied to Chandrayaan-2 by the ISDA Archive Working Group. In addition to these documents, the Chandrayaan-2 Archive Working Group has produced Chandrayaan-2 Archive conventions containing the additional XML attribute and parameter definitions specific to Chandrayaan-2 Orbiter mission.

Examples of PDS labels required for the OHRC archive are shown in Appendix B (bundle products), Appendix C (collection products), Appendix D (basic products), Appendix E (Data visualization using PDS4 Viewer), Appendix F (Count to radiance Conversion) and Appendix G (Local Data Dictionary).

7.4 Delivery Package

Data transfers, from ISDA to the data users, are accomplished using delivery packages. Delivery packages include the following required elements:

- The package which consists of a compressed bundle of the products being transferred.
- A transfer manifest which maps each product's LID/VID to the physical location of the product label in the package after uncompressing.
- A checksum manifest which lists the MD5 checksum of each file included in the package after uncompressing.

Appendix A Support staff and cognizant Persons

Table 24 Chandrayaan-2 Team staff

Teams	Person Name	Contact	Email Id
Operations Team @ ISSDC IDSN Bylalu, Bengaluru	Sri. Sreenath	080-2809-4416/17/18 (SCC), 2809-4419 (MOX), 2202-9173/74 (ISSDC-IDSN)	sreenath@istrac.gov.in
Principal Investigator (PI) Team @ SAC Ahmedabad	Aditya Dagar	07926914309	adagar@sac.isro.gov.in

Appendix B Sample Bundle Product Label

```

<Product_Bundle xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
https://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1700.xsd">
  <Identification_Area>
    <logical_identifier>urn:isro:isda:ch2_cho</logical_identifier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-2 OHRC Data Archive Products Bundle</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Bundle</product_class>
    <Citation_Information>
      <author_list>isro,issdc,sac</author_list>
      <publication_year>2020</publication_year>
      <description>Chandrayaan-2 Orbiter OHRC Payload Data Products</description>
    </Citation_Information>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2020-12-02</modification_date>
        <version_id>1.0</version_id>
        <description>Peer reviewed version of the Chandrayaan-2 orbiter OHRC payload
data products bundle</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Context_Area>
    <Time_Coordinates>
      <start_date_time>2020-02-29Z</start_date_time>
      <stop_date_time>2020-03-02Z</stop_date_time>
    </Time_Coordinates>
    <Investigation_Area>
      <name>Chandrayaan-2</name>
      <type>Mission</type>
      <Internal_Reference>
        <lid_reference>urn:isro:isda:ch2_cho.ohr:context:mision</lid_reference>
        <reference_type>bundle_to_investigation</reference_type>
      </Internal_Reference>
    </Investigation_Area>
    <Observing_System>
      <name>Chandrayaan-2</name>
      <Observing_System_Component>
        <name>Chandrayaan-2 Orbiter</name>
        <type>Spacecraft</type>
        <Internal_Reference>
          <lid_reference>urn:isro:isda:ch2_cho.ohr:context:inst_host_spacecraft</lid_referenc
e>

```

```

    <reference_type>is_instrument_host</reference_type>
  </Internal_Reference>
</Observing_System_Component>
<Observing_System_Component>
  <name>orbiter high resolution camera</name>
  <type>Instrument</type>
  <Internal_Reference>
    <lid_reference>urn:isro:isda:ch2_cho.ohr:context:inst_ohr</lid_reference>
    <reference_type>is_instrument</reference_type>
  </Internal_Reference>
</Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <Internal_Reference>
    <lid_reference>urn:isro:isda:context:target:satellite.moon</lid_reference>
    <reference_type>data_to_target</reference_type>
  </Internal_Reference>
</Target_Identification>
</Context_Area>
<Bundle>
  <bundle_type>Archive</bundle_type>
</Bundle>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:calibration</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_calibration_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:data</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_data_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:document</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_document_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:context</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_context_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:browse</lid_reference>
  <member_status>Primary</member_status>

```

```
<reference_type>bundle_has_browse_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:geometry</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_geometry_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:miscellaneous</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_member_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.ohr:schema</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_schema_collection</reference_type>
</Bundle_Member_Entry>
</Product_Bundle>
```

Appendix C Sample Collection Product Label

```

<Product_Collection xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1400.xsd
http://pds.nasa.gov/pds4/disp/v1
http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1
http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd">

  <Identification_Area>
    <logical_identifier>urn:isro:isda:ch2_cho.ohr:document</logical_identifier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-2 Orbiter OHRC document file collection</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Collection</product_class>
    <Citation_Information>
      <author_list>Chandrayaan-2 DP Team</author_list>
      <editor_list>Chandrayaan-2 DP Team</editor_list>
      <publication_year>2020</publication_year>
      <description>collection of document products.</description>
    </Citation_Information>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2020-12-02</modification_date>
        <version_id>1.0</version_id>
        <description>Initial version</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Context_Area>
    <Investigation_Area>
      <name>chandrayaan-2</name>
      <type>Mission</type>
      <Internal_Reference>

<lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan2::1.0</lid
vid_reference>
      <reference_type>collection_to_investigation</reference_type>
    </Internal_Reference>
    </Investigation_Area>
    <Observing_System>
      <Observing_System_Component>
        <name>Chandrayaan 2 Orbiter</name>
        <type>Spacecraft</type>
        <description>

```

Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

```

</description>
</Observing_System_Component>
<Observing_System_Component>
  <name>orbiter high resolution camera</name>
  <type>Instrument</type>
  <description>
    OHRC (Orbiter High Resolution Camera) is one of the scientific instrument hosted
    on the Chandrayaan-2 Orbiter. The instrument is a a passive electro-optical
    imaging camera operating in visible-panchromatic band (500-800nm).
    OHRC images the lunar surface with a ground sampling distance of 0.25m
  </description>
</Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <description>Moon is a natural satellite of Earth</description>
</Target_Identification>
</Context_Area>
<Collection>
  <collection_type>Document</collection_type>
</Collection>
<File_Area_Inventory>
  <File>
    <file_name>collection_document_inventory.csv</file_name>
    <local_identifier>collection_document_inventory.csv</local_identifier>
    <creation_date_time>2020-07-28</creation_date_time>
    <file_size unit="byte">73</file_size>
    <records>1</records>
    <md5_checksum>09a718293d7ee74abaafa2722619865f</md5_checksum>
  </File>
<Inventory>
  <offset unit="byte">0</offset>
  <parsing_standard_id>PDS DSV 1</parsing_standard_id>
  <records>1</records>
  <record_delimiter>Carriage-Return Line-Feed</record_delimiter>
  <field_delimiter>Comma</field_delimiter>
  <Record_Delimited>
    <fields>2</fields>
    <groups>0</groups>
    <maximum_record_length unit="byte">259</maximum_record_length>
  </Field_Delimited>

```

```

    <name>Member Status</name>
    <field_number>1</field_number>
    <data_type>ASCII_String</data_type>
    <maximum_field_length unit="byte">1</maximum_field_length>
    <description>This column specifies the member status</description>
  </Field_Delimited>
  <Field_Delimited>
    <name>LIDVID_LID</name>
    <field_number>2</field_number>
    <data_type>ASCII_LIDVID_LID</data_type>
    <maximum_field_length unit="byte">255</maximum_field_length>
    <description>This column specifies the LIDVID of the files that comprise the
collection.</description>
  </Field_Delimited>
</Record_Delimited>
  <reference_type>inventory_has_member_product</reference_type>
</Inventory>
</File_Area_Inventory>

</Product_Collection>

```

Appendix D Sample Data Product Labels

```

<Product_Observational xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1400.xsd
http://pds.nasa.gov/pds4/disp/v1
http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1
http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd">
  <Identification_Area>
    <logical_idenfier>urn:isro:isda:ch2_cho:ohr_calibrated:ch2_ohr_ncp_20200302t01
05126060_d_img_g26</logical_idenfier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-2 Orbiter OHRC Experiment</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Observational</product_class>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2017-12-22</modification_date>
        <version_id>1.0</version_id>
        <description>PDS4 product label created by SAC Optical DP Team</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Observation_Area>
    <Time_Coordinates>
      <start_date_time>2020-03-02T01:05:12.6060Z</start_date_time>
      <stop_date_time>2020-03-02T01:05:28.9896Z</stop_date_time>
    </Time_Coordinates>
    <Primary_Result_Summary>
      <purpose>Science</purpose>
      <processing_level>Calibrated</processing_level>
    </Primary_Result_Summary>
    <Investigation_Area>
      <name>Chandrayaan-2</name>
      <type>Mission</type>
      <Internal_Reference>
        <lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan2::1.0</lid
vid_reference>
        <reference_type>data_to_investigation</reference_type>
      </Internal_Reference>
    </Investigation_Area>
    <Observing_System>
      <Observing_System_Component>
        <name>Chandrayaan 2 Orbiter</name>
        <type>Spacecraft</type>
        <description>

```

Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

```

</description>
</Observing_System_Component>
<Observing_System_Component>
  <name>orbiter high resolution camera</name>
  <type>Instrument</type>
  <description>
    OHRC (Orbiter High Resolution Camera) is one of the scientific instrument hosted
    on the Chandrayaan-2 Orbiter. The instrument is a passive electro-optical
    imaging camera operating in visible-panchromatic band (500-800nm).
    OHRC images the lunar surface with a ground sampling distance of 0.25m
  </description>
</Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <description>Moon is a natural satellite of Earth</description>
</Target_Identification>
<Mission_Area>
  <Product_Parameters>
    <job_id>OHRXXG26CHO0232002NNNN20062031920459_V1_0</job_id>
<level0_dir_name>OHRXXG26CHO0232002NNNN20062031920459_V1_0</level0_dir_name>
    <imaging_orbit_number>2318</imaging_orbit_number>
    <dumping_orbit_number>2320</dumping_orbit_number>
    <line_exposure_duration unit="ms">162.100</line_exposure_duration>
    <bits_selection>msb</bits_selection>
    <tdi_stages>TDI64</tdi_stages>
    <detector_pixel_width unit="micron">5.2</detector_pixel_width>
    <focal_length unit="mm">2080</focal_length>
    <spacecraft_altitude unit="km">90.650125789544319</spacecraft_altitude>
    HORIZONTAL_PIXEL_SCALE = -- m/pixel
    VERTICAL_PIXEL_SCALE == -- m/pixel
    RESOLUTION == -- m/pixel
    <reference_data_used>SELENE</reference_data_used>
    <orbit_limb_direction>Ascending</orbit_limb_direction>
    <spacecraft_yaw_direction>True</spacecraft_yaw_direction>
  </Product_Parameters>
  <Geometry_Parameters>
    <System_Level_Coordinates>
      <upper_left_latitude unit="deg">-70.881680</upper_left_latitude>
      <upper_left_longitude unit="deg">23.197066</upper_left_longitude>
      <upper_right_latitude unit="deg">-70.877277</upper_right_latitude>
      <upper_right_longitude unit="deg">22.895169</upper_right_longitude>
      <lower_left_latitude unit="deg">-70.039155</lower_left_latitude>

```

```

    <lower_left_longitude unit="deg">23.291884</lower_left_longitude>
    <lower_right_latitude unit="deg">-70.034940</lower_right_latitude>
    <lower_right_longitude unit="deg">23.002551</lower_right_longitude>
  </System_Level_Coordinates>
  <Refined_Corner_Coordinates>
    <upper_left_latitude unit="deg">-71.018205</upper_left_latitude>
    <upper_left_longitude unit="deg">23.219009</upper_left_longitude>
    <upper_right_latitude unit="deg">-71.014050</upper_right_latitude>
    <upper_right_longitude unit="deg">22.921632</upper_right_longitude>
    <lower_left_latitude unit="deg">-70.175621</lower_left_latitude>
    <lower_left_longitude unit="deg">23.312214</lower_left_longitude>
    <lower_right_latitude unit="deg">-70.171791</lower_right_latitude>
    <lower_right_longitude unit="deg">23.027298</lower_right_longitude>
  </Refined_Corner_Coordinates>
</Geometry_Parameters>
</Mission_Area>
</Observation_Area>
<File_Area_Observational>
  <File>
    <file_name>ch2_ohr_ncp_20200302T0105126060_d_img_g26.img</file_name>
    <creation_date_time>2020-07-28T04:32:35</creation_date_time>
    <file_size unit="byte">1212900000</file_size>
    <md5_checksum>36ff87c8f4766f9bf6b9ce8b92144cf3</md5_checksum>
    <comment>
      This File contains the count calibrated image
      md5_checksum is provided for ensuring data integrity when users are
      downloading the data.
    </comment>
  </File>
  <Array_2D_Image>
    <offset unit="byte">0</offset>
    <axes>2</axes>
    <axis_index_order>Last Index Fastest</axis_index_order>
    <Element_Array>
      <data_type>UnsignedByte</data_type>
    </Element_Array>
    <Axis_Array>
      <axis_name>Line</axis_name>
      <elements>101075</elements>
      <sequence_number>1</sequence_number>
    </Axis_Array>
    <Axis_Array>
      <axis_name>Sample</axis_name>
      <elements>12000</elements>
      <sequence_number>2</sequence_number>
    </Axis_Array>
  </Array_2D_Image>

```

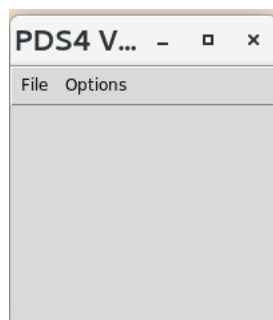
</File_Area_Observational>
</Product_Observational>

Appendix E Visualize PDS4 Data Products using PDS4 Viewer

There are many PDS tools are available for opening the Raw/Calibrated Data products. PDS4Viewer - PDS4Viewer is a simple display tool for visualizing data contained in the PDS4 data standard. PDS4Viewer tool can be downloaded from internet using link "http://sbndev.astro.umd.edu/wiki/PDS4_Viewer". This Software is available for both Linux as well as window platform. The given data sets can be opened using PDS4Viewer.

The instructions to open the data using PDS4Viewer are as follows:

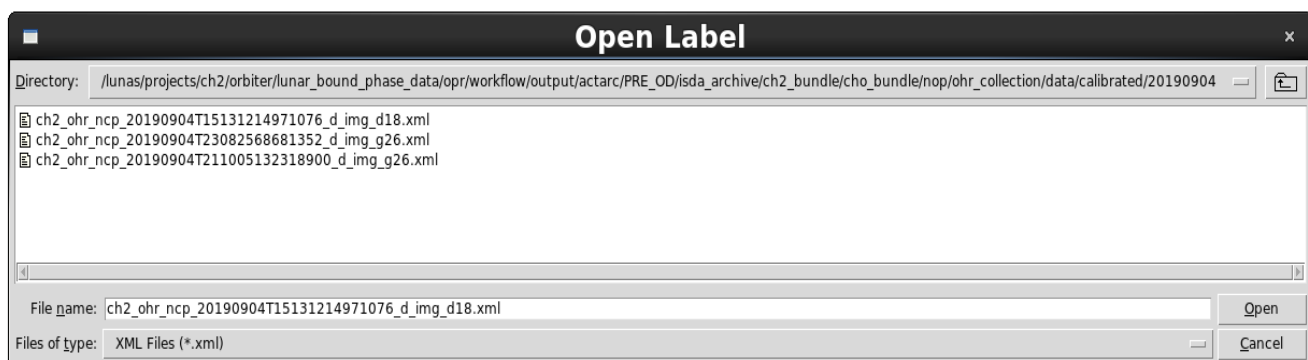
1. Download the PDS4Viewer tool from internet http://sbndev.astro.umd.edu/wiki/PDS4_Viewer. There will be a zip file (for both Linux/Windows) contains the PDS4Viewer.
2. Extract the Zip file into any location.
3. Go to extract the directory, run pds4_viewer. The PDS4Viewer GUI will come which is shown below.



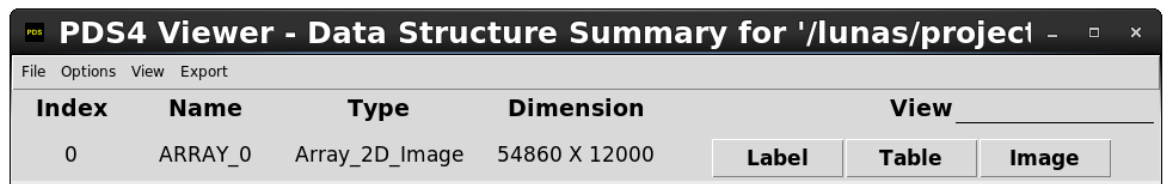
Opening PDS4 Raw/Calibrated Image Product

=====

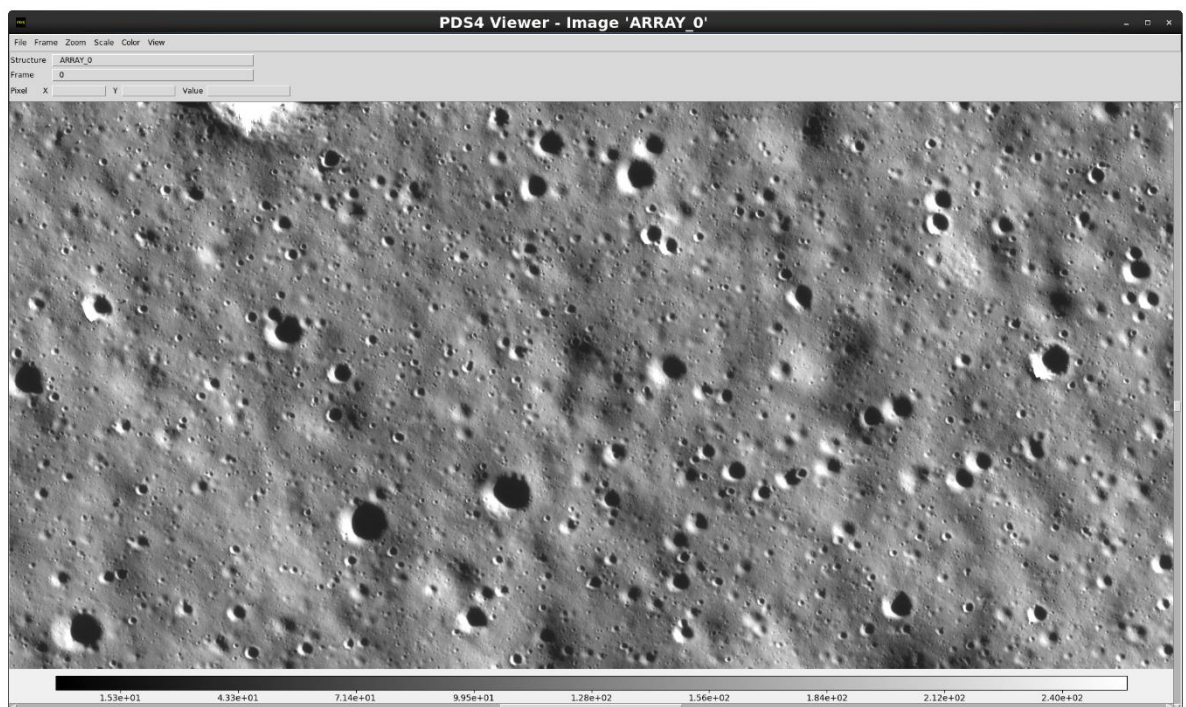
1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



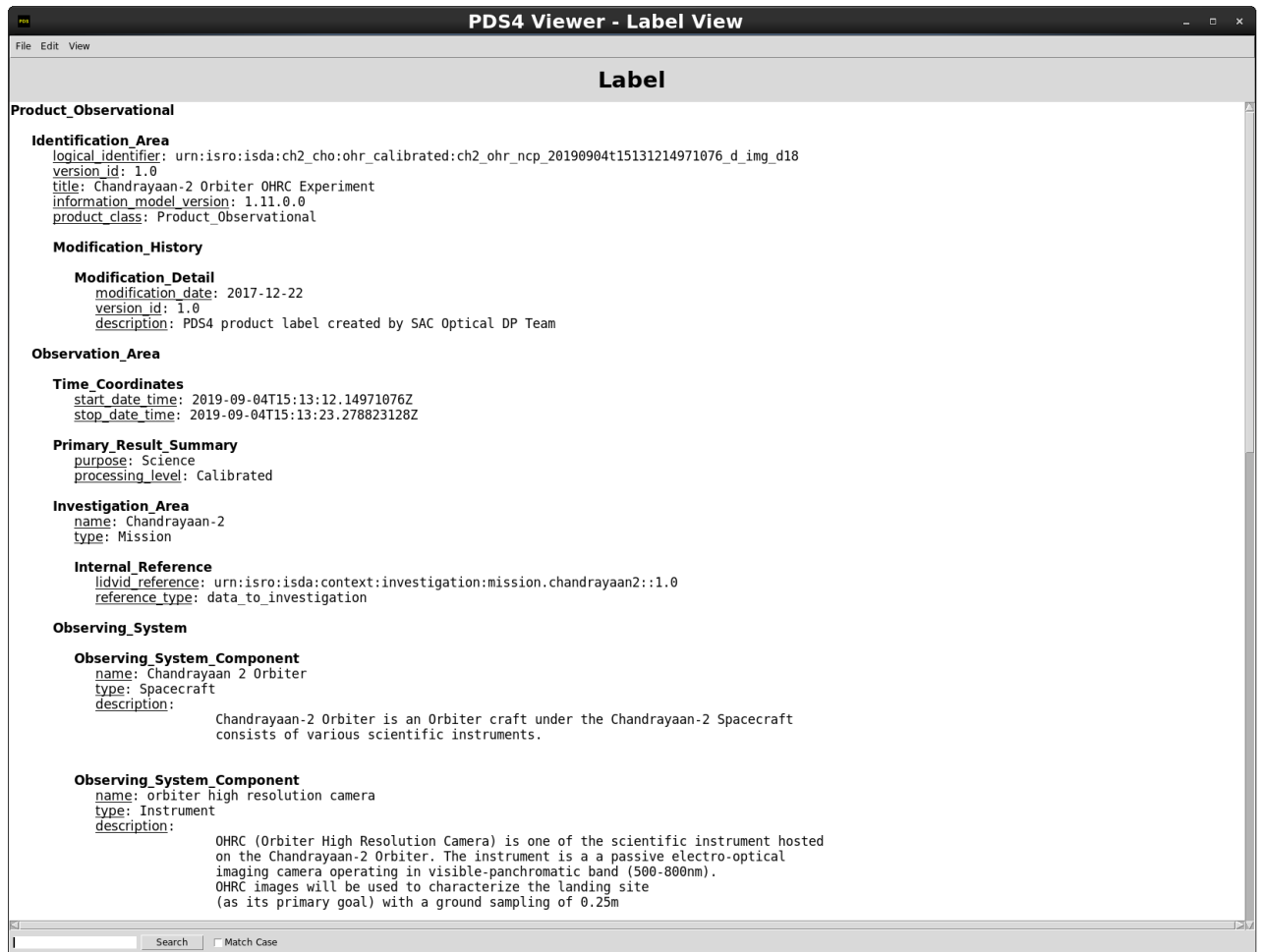
3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:



4. For Opening an image, click on the Image button shown on the above figure at step no. 3. Below shows the image from the OHRC Payloads during Decision Support System Operations at MOX on 04 September 2019.



5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view.



Opening PDS4 Geometry Grid Data Product

=====

1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



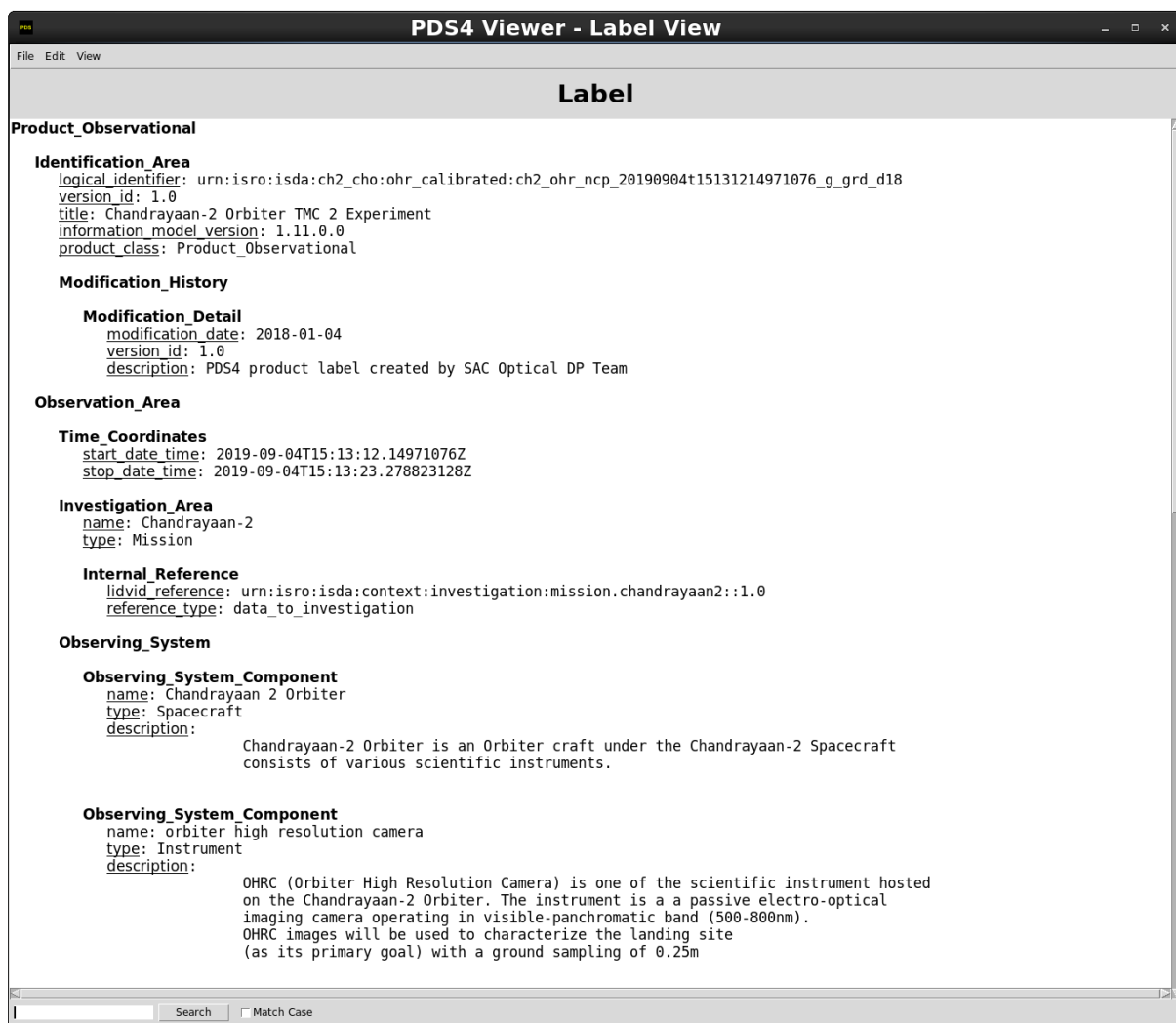
3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:

Index	Name	Type	Dimension	View
1	STREAM_CSV_ID	Table_Delimited	4 cols X 66550 rows	Label Table Plot

4. For Opening grid file, click on the table button shown on the above figure at step no. 3. Below shows the grid file from the OHRC Payloads

Row #	Longitude	Latitude	Pixel	Scan
0	4.184324	-74.2995049	0	0
1	54.1876176	-74.2999071	100	0
2	54.1909109	-74.3003091	200	0
3	54.1942041	-74.3007111	300	0
4	54.1974971	-74.301113	400	0
5	54.2007899	-74.3015148	500	0
6	54.2040825	-74.3019164	600	0
7	54.207375	-74.302318	700	0
8	54.2106672	-74.3027195	800	0
9	54.2139593	-74.3031209	900	0
10	54.2172512	-74.3035221	1000	0
11	54.2205429	-74.3039233	1100	0
12	54.2238345	-74.3043244	1200	0
13	54.2271258	-74.3047254	1300	0
14	54.230417	-74.3051262	1400	0
15	54.2337079	-74.305527	1500	0
16	54.2369987	-74.3059277	1600	0
17	54.2402894	-74.3063283	1700	0
18	54.2435798	-74.3067287	1800	0
19	54.24687	-74.3071291	1900	0
20	54.2501601	-74.3075294	2000	0
21	54.25345	-74.3079295	2100	0
22	54.2567397	-74.3083296	2200	0
23	54.2600292	-74.3087296	2300	0
24	54.2633185	-74.3091295	2400	0
25	54.2666077	-74.3095292	2500	0
26	54.2698966	-74.3099289	2600	0
27	54.2731854	-74.3103285	2700	0
28	54.276474	-74.310728	2800	0
29	54.2797624	-74.3111274	2900	0
30	54.2830507	-74.3115266	3000	0
31	54.2863387	-74.3119258	3100	0
32	54.2896266	-74.3123249	3200	0
33	54.2929143	-74.3127239	3300	0
34	54.2962018	-74.3131227	3400	0
35	54.2994891	-74.3135215	3500	0
36	54.3027762	-74.3139202	3600	0
37	54.3060632	-74.3143188	3700	0
38	54.30935	-74.3147173	3800	0
39	54.3126366	-74.3151157	3900	0
40	54.315923	-74.3155139	4000	0
41	54.3192092	-74.3159121	4100	0
42	54.3224952	-74.3163102	4200	0

5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view



Appendix F Count to Radiance Conversion

The OHRC Raw and Calibrated image data products are the count data products in order to convert to radiance data product. Following formula is used for the conversion

1. Formula for Conversion of Count to Radiance

$$\text{radiance value} = \frac{(\text{Max_SR} - \text{Min_SR}) * \text{gray_value}}{\text{no_of_gray_value}}$$

Note:

- SR stands for Saturation Radiance.
- number_of_gray_value will be calculated as 2^n , where n is quantization bit (for ohrc it is 8 bit).
- Min_SR and Max_SR are the minimum Saturation and maximum Radiances respectively.
- Min_SR value is 0
- In OHRC, there are various TDI stages - 64, 128, 256. Max_SR value can be obtained from the table

Table 25 shows the saturation radiance information for OHRC

Table 25 OHRC Saturation Radiance

OHRC	TDI Stages	Min_SR	Max_SR
PAN	64	0	3.87
	256	0	0.8

Appendix G Local Data Dictionary

Every mission has some conditions based on the events that needs to be handle as an information at output product level. Some information is common to every planetary mission launched by ISRO. There is an ISRO Science Data Archive established during Chandrayaan-1 mission in 2008. All the products hosted for users under ISDA are PDS compliance. For Chandrayaan-2 mission, PDS4 Archive standard was adopted. PDS4 is currently next and future generation archive standard adopted by all the space agencies including NASA, ESA and JAXA for their current and future mission. To handle the ISRO planetary mission's specific parameters, there is a concept in PDS4 called Local Data Dictionary i.e. LDD. In LDD one can define all the parameters and that can be used as a reference for future planetary missions in India. LDD takes basic data dictionary of PDS4 and then top of it builds local dictionary. For Chandrayaan-2 mission and future planetary missions of ISRO, list of parameters is identified shown in details in table 26.

Table 26 ISDA Local Data Dictionary Class and Attributes List

Class/Sub Class Hierarchy	Attribute Name	Description/Value
Mission_Area Product_Parameters	job_id	Unique id assign by DP Scheduler when the product is ingested
	level0_dir_name	DP Input Level-0 directory name
	imaging_orbit_number	Imaging Orbit Number assigned on board
	dumping_orbit_number	Dumping Orbit Number is assigned based on the data collected on ground station
	line_exposure_duration	Image integration time
	bits_selection	Bit selection based on TDI settings. It can have values like lsb, msb, mid
	tdi_stages	Time delay integration stages
	detector_pixel_width	Pixel width of detector
	focal_length	Instrument focal length
	spacecraft_altitude	Spacecraft height
	orbit_limb_direction	Orbit limb direction it can have two values - Ascending or Descending Spacecraft yaw mode it can have two values – True/False
	spacecraft_yaw_direction	

		a) True means Reverse, in flight yaw mode Reverse describes 180 degree rotated from nominal. b) False means Forward, in flight yaw mode Forward describes 0 degree rotated from nominal) There are four possible orbit_limb_direction and spacecraft_yaw_direction combinations: 1. Descending and False, no changes in lines or samples are performed 2. Descending and True, samples are reversed 3. Ascending and False, Lines are reversed 4. Ascending and True. both lines and samples are reversed
	reference_data_used	Reference data set used in refining the system level corners List of Possible Values 1. SELENE (10 meter ortho reference) 2. LRO (100 meter ortho reference) 3. CLEMENTINE (100 meter ortho reference) 4. System (available reference but no match found gone with system level) 5. n/a means not available
	pixel_resolution	Image pixel resolution expressed in unit m/pixel
	roll	Spacecraft spinning on X-axis value expressed in degree
	pitch	Spacecraft spinning on Y-axis value expressed in degree
	yaw	Spacecraft spinning on Z-axis value expressed in degree
	sun_azimuth	Angle between the projection of sun rays and a

		line due south or north and value is expressed in degree
	sun_elevation	Angle between the horizon and the Sun's disc centre and value is expressed in degree
	solar_incidence	Angle between the sun's rays and the normal on a surface and value is expressed in degree
	projection	projection is map projection which is set of transformations employed to represent the curved two-dimensional surface of a globe on plane. List of possible values 1. Selenographic 2. Polar stereographic
	area	Area on which imaging is done. List of Possible Values 1. Equatorial 2. North Pole 3. South Pole
Mission_Area - Geometry_Parameters - System_Level_Coordinates - Refined_Level_Coordinates	upper_left_latitude	Upper left latitude reticle point with number 1
	upper_left_longitude	Upper left longitude reticle point with number 1
	upper_right_latitude	Upper right latitude reticle point with number 3
	upper_right_longitude	Upper right longitude reticle point with number 3
	lower_left_latitude	Lower left latitude reticle point with number 7
	lower_left_longitude	Lower left longitude reticle point with number 7
	lower_right_latitude	Lower right latitude reticle point with number 9
	lower_right_longitude	Lower right longitude reticle point with number 9

Appendix H Data Disclaimer

Orbiter High Resolution Camera (OHRC) is a very high spatial resolution (0.25 m at nadir from 100 km orbit) imaging optical panchromatic camera on-board Chandrayaan-2. This instrument is primarily designed for imaging of small surface features under low illumination conditions.

The public release of OHRC data products has two levels- Level-0 (Raw) and Level-1 (Calibrated).

Level-0 data is the un-calibrated data in count numbers (8-bit unsigned integer, generic binary format). It can be converted to level-1 data by using the equation along with the maximum & minimum saturation radiances provided in Appendix F.

Level-1 is radiometric calibrated data (8-bit unsigned integer, generic binary format) with seleno tagged information. Ancillary information is provided as text files.

OHRC Level-1 datasets generated may have these limitations:

There may be slight deviation in the corner coordinates due to the pre orbit determination.

OHRC works on TDI technology and at higher TDI level some of the portion of OHRC data may contain saturation (DN=255) due to higher radiance encountered by the camera.

The following datasets have been corrected up to the maximum possible extent. There is still some noise present in ports 2000-2399,3600-3999 and 8400-8799 which may need further correction before actual data interpretation.

- PDS4 Raw Data Sets:
 ch2_ohr_nrp_20240316T2008014680_d_img_d18
 ch2_ohr_nrp_20240316T2205443428_d_img_d18
- PDS4 Calibrated Data Sets:
 ch2_ohr_ncp_20240316T2008014680_d_img_d18
 ch2_ohr_ncp_20240316T2205443428_d_img_d18

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